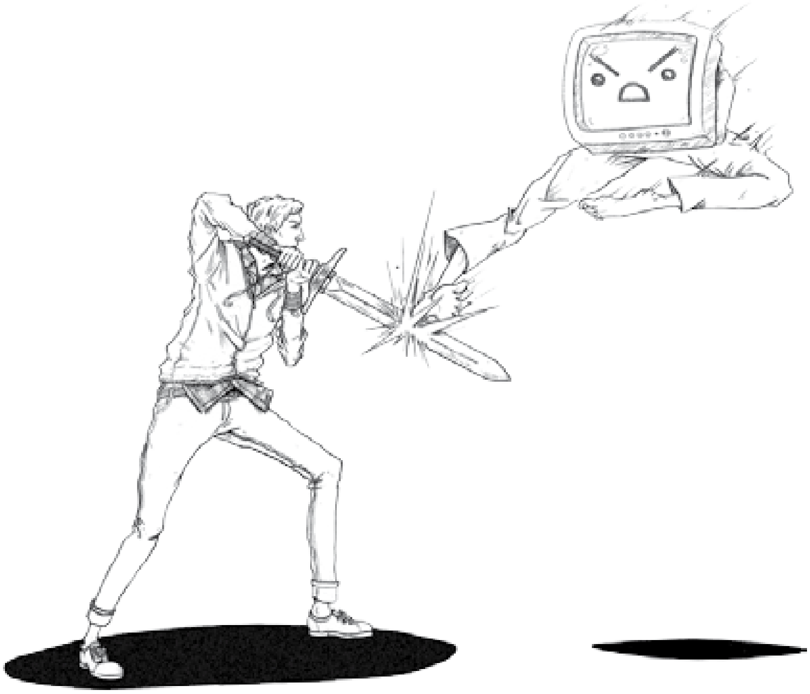
gies that enable such interpretation will transform the human-computer interaction.



ENGINEERING CHALLENGES FOR A NEW WORLD

Today, most users communicate with their computers using the familiar keyboard and mouse, which offer a direct and recognizable collection of input for computers and present simple data points for the software environment to evaluate. When a user presses a key or clicks a mouse button, little about those actions can be misconstrued or misinterpreted. However, these actions limit users to a single interface.

Developers are now working to make computers as cognitive about their surroundings as we are (or at least as aware as we should be) so they can process much of the information around them and arrive at a logical conclusion based on a user's intent. Extracting information from the environment will include data points such as the directionality of the voice (for example, is the user talking to the computer or to a friend nearby?), the ambient background noise, facial recognition for automatic user selection (and security), 3D maps of the environment for object and gesture recognition, and more.

To reach that goal, many important engineering tasks must be addressed, starting with the integration of new and improved sensors on PCs, Ultrabook™ devices, tablets, and smartphones. Cameras that can evaluate depth, microphones that understand directional audio, and touchpads with pressure sensitivity need to be standardized, miniaturized, and implemented across the ecosystem. For perceptual computing to take a firm hold, devices must be equipped with the next level of intelligence and capabilities.

Advances in environment awareness are becoming enabled by higher performance computing platforms, including the 3rd generation Intel® Core™ processor family found in Ultrabook devices. Intel engineers have developed improved process technology and CPUs that can maintain a real-time connection with different communication interfaces and can provide a high level of user experience. Although many hardware challenges remain—including how to address enormous data sets and further miniaturize sensors—Moore's Law and Intel will continue to unlock the power of perceptual computing.

Perceptual computing is not about changing and remapping current interfaces, so don't expect the keyboard and mouse to vanish from the world of PCs any time soon. Perceptual computing aims instead to create new modalities that redefine computing interactions.